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Locality preserving and homomorphically encrypted deep features for privacy-preserving face recognition

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Abstract

Face recognition (FR) has emerged as a vital biometric authentication method due to its non-invasive nature and widespread applications in security, surveillance, and identity verification. However, existing FR systems face significant challenges in preserving user privacy, especially when deployed in cloud environments. Traditional approaches lack robust encryption mechanisms, making sensitive facial data vulnerable to breaches, adversarial attacks, and unauthorised access. This study introduces a novel privacy-preserving face recognition (PPFR) framework that leverages deep learning (DL) models, specifically FaceNet and ArcFace, for extracting discriminative facial embeddings. The framework combines these models with the Cheon–Kim–Kim–Song (CKKS) homomorphic encryption scheme to enable secure computation on encrypted data without requiring decryption. Locality Preserving Projection (LPP) is applied to compress high-dimensional features while preserving the essential data structure, thereby enhancing performance and reducing processing overhead. The proposed system was evaluated on four benchmark datasets: CASIA3D, 105PinsFace, LFW, and Faces94. Among the models, ArcFace showed the best performance, achieving high accuracy and low Equal Error Rates (EER), with results of 99.30% accuracy on CASIA3D, 98.06% on 105PinsFace, 97.07% on LFW, and 99.82% on Faces94 datasets. The experimental results demonstrate that the proposed system is both effective and secure, making it suitable for real-world face recognition applications with enhanced privacy protection.

Keywords Face recognition, Privacy preserving, Facenet, Arcface, Facial embeddings, CKKS homomorphic encryption

1 Introduction

As technology becomes increasingly integrated into everyday life, having a secure, user-friendly way to verify a person's identity is more important than ever. Traditional methods, such as passwords, PINs, and security tokens, are still commonly used; however, they come with several drawbacks. Passwords may be weak, reused across platforms, or forgotten, while tokens can be lost or stolen. These issues underline the need for more secure and dependable authentication solutions. Biometric authentication offers

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a stronger alternative by using unique physical traits—such as fingerprints, iris patterns, voice, or facial features—to verify identity [1]. Since these traits are unique to each person, they are generally harder to steal or forge compared to traditional methods [2]. Among various biometric methods, FR stands out due to its ease of use, non-intrusive nature, and high accuracy [3]. The widespread availability of high-resolution cameras and the rapid advancement of DL-based FR algorithms have enabled the deployment of FR systems, particularly on personal devices such as smartphones and laptops [4]. However, despite their popularity, current FR systems still face several significant challenges. These include vulnerability to spoofing attacks [5], potential misuse of facial data if compromised, and the inability to change one's biometric traits once leaked. Additionally, centralised databases of facial templates present attractive targets for attackers [6].

A typical biometric system consists of two main phases: enrolment and authentication. During enrolment, biometric data is captured and processed using a feature extractor to identify key characteristics. These features are used to create a reference template, which is securely stored. During the authentication phase, new biometric data is captured, and a query template is generated using the same feature extractor. This template is then compared to the stored reference. If the templates match, access is granted; otherwise, it is denied. The reliability of this process depends heavily on the quality of the feature extractor and the secure handling of sensitive data. As more facial recognition (FR) systems migrate to cloud environments, protecting facial data becomes a greater concern [7]. Cloud platforms offer strong computing power and storage, but they also create more chances for data leaks or misuse. While biometrics help reduce fraud, keeping templates safe from hackers remains a significant challenge [8]. To tackle this issue, researchers have proposed various biometric template protection schemes, including cancellable biometrics [9], bio-cryptosystems [10], and homomorphic encryption (HE) [11]. HE, in particular, enables computations on encrypted data without requiring decryption, offering strong privacy guarantees. Based on their capabilities, HE schemes can be classified as Partially Homomorphic Encryption (PHE) [12], Somewhat Homomorphic Encryption (SHE) [13], and Fully Homomorphic Encryption (FHE) [14], with FHE being the most powerful and suitable for secure biometric computations.

1.1 Motivation

Most privacy-preserving face recognition methods try to protect users by encrypting the entire high-dimensional face feature vectors produced by deep neural networks. However, these vectors are usually large; encrypting such large data with homomorphic encryption produces large ciphertexts and slows matching, which is a problem for real-time or low-resource systems. Even though face recognition technology has improved significantly, many systems still do not keep data encrypted from end to end. At some point in the pipeline, sensitive features may be exposed in plain form, creating privacy risks. This shows the need for a practical system that fully protects privacy without sacrificing accuracy or usability. To solve this, the proposed system uses a deep-learning model together with the CKKS homomorphic encryption scheme. CKKS can perform arithmetic operations on encrypted real-valued data, making it suitable for working with feature vectors from CNNs. By keeping the data encrypted throughout the entire process, the system aims to provide strong privacy protection while still maintaining reliable face recognition performance.

1.2 Objectives

- To build a privacy-aware face recognition system based on deep learning and homomorphic encryption.
- To implement Locality Preserving Projection (LPP) of dimensionality reduction of deep face embedding to reduce the dimensions, To compare the effects of various template sizes (e.g., 512-D and smaller dimensions) on the recognition accuracy and the cost of encrypted computation.
- To create face matching at the encrypted domain without exposing the face features.
- To minimize the size of the ciphertext and the computation time, but still have a reliable performance on recognition.
- To test the proposed framework with a number of data sets in different conditions.

1.3 Contributions

The proposed system offers a secure and reliable face recognition system. The contributions of the proposed system are as follows:

- A privacy-preserving face recognition framework is developed by integrating FaceNet/ArcFace embeddings, Locality Preserving Projection (LPP), and CKKS homomorphic encryption to enable encrypted feature matching without revealing facial information.
- LPP is applied to deep facial embeddings to reduce feature dimensionality from 512-D to 256-D while maintaining local geometric structure. This reduction reduces ciphertext size and improves the efficiency of encrypted computation with minimal impact on recognition accuracy.
- CKKS encryption parameters are systematically analysed and optimised to achieve an effective balance between precision, security level, and computational cost.
- The proposed framework is evaluated on four diverse datasets covering both controlled and uncontrolled settings, demonstrating strong adaptability and robustness under varying conditions.
- A comprehensive ablation study is conducted to compare encrypted matching using 512-D and 256-D embeddings, showing that LPP significantly enhances computational efficiency while maintaining reliable recognition performance.

The rest of the paper is structured as follows: Section 2 reviews related work on DL-based privacy-preserving face recognition. Section 3 explains our proposed FHE-based method. Section 4 presents results and comparisons. Section 5 concludes with key findings and future directions.

2 Related work

Face recognition (FR) is widely used across security, finance, and healthcare. Many deep learning methods, especially CNN-based models, have improved accuracy under challenging conditions such as poor lighting and occlusions. However, most focus only on performance and ignore privacy concerns. Since biometric data is sensitive, there is a growing need for privacy-preserving methods. Techniques like HE allow secure processing without exposing the data. While some privacy-focused FR systems exist, achieving both high accuracy and strong data protection is still difficult. This challenge has

motivated our work on a secure biometric authentication system. A summary of related works, including key methods and Limitations, is provided in the Table 1

Chouchane et al. [16] proposed the HOSVD+EFM method. It utilises MB-LPQ features from depth images, which are shaped as third-order tensors. They applied HOSVD with an Enhanced Fisher Model (EFM). This method achieved 98.60% accuracy on the CASIA-3D dataset. It worked without pose-specific training. Bessaoudi et al. [19] introduced MEFDA. It supports multilinear discriminant analysis and preserves the structure of tensor data. DL has recently shown great promise in biometric systems. Convolutional neural networks (CNNs) are widely used and successful. Tamilselvi et al. [24] proposed the HRPSM_CNN model. It helps face recognition in poor lighting and occlusion. This model is designed for real-time applications, such as assistive devices. Abuzneid et al. [18] used Back Propagation Neural Networks (BPNN) with LBPH descriptors and multi-KNN. They improved training using preprocessing and LBP-based local feature extraction. Pranav et al. [20] used a CNN model. It achieved 98.75% accuracy on the ORL dataset.

Aliradi et al. [36] introduced the LGBQ descriptor. It combines Gabor filters, LBP, and LPQ features. Chen et al. [30] used CNNs to convert 2D images into depth maps. They achieved 96.75% accuracy on CASIA-3D. This method helped expand DL use in biometric tasks. Li et al. Li et al. [38] developed a disentangled representation transformer network for 3D face reconstruction and robust dense alignment. Their approach separates identity and expression features, improving the network's robustness to occlusions and

Table 1 Summary of related work with methods and limitations

Author(s)	Method/technique	Limitations
Ma et al. [15]	DL with HE for secure verification	Requires more scalability testing
Chouchane et al. [16]	HOSVD + EFM with MB-LPQ features	No pose-specific training; lacks privacy focus
Boddeti et al. [17]	Batching and dimensionality reduction with FHE	Resource-intensive despite optimizations
Abuzneid et al. [18]	LBPH + BPNN + multi-KNN	Improved robustness; lacks privacy
Bessaoudi et al. [19]	MEFDA for tensor structure preservation	Does not address privacy concerns
Pranav et al. [20]	CNN-based recognition	Limited dataset
Ali et al. [21]	SVM with SqueezeNet	Moderate accuracy; no privacy
Miakshyn et al. [22]	FaceNet + OpenFace	Low accuracy; lacks privacy features
Khan et al. [23]	Encrypted FR with Local Ternary Patterns	Limited to handcrafted descriptors
Tamilselvi et al. [24]	HRPSM_CNN for robust recognition	No security layer; focused on real-time use
Alirezazadeh et al. [25]	ACML + DenseNet121	Strong results; limited scalability analysis
Albraikan et al. [26]	ResNet152v2 with transfer learning	Task-specific; lacks privacy protection
Lee et al. [27]	FHE with DL-based FR	Secure but computationally heavy
Ke et al. [28]	LocalFace with facial attention	Works with occlusions; lacks privacy
Huang et al. [29]	Lightweight feature restoration network	Enhances visibility; no privacy model
Chen et al. [30]	CNN to convert 2D to depth map	Effective, but no secure processing
Cheema et al. [31]	DNDR with joint loss	Handles disguise; complex training
Xie et al. [32]	Local differential privacy for edge-based FR	Lightweight; no deep model integration
Yang et al. [33]	HeadPose-Softmax with curriculum learning	Pose handling; lacks encryption
Mi et al. [34]	FHE with DL for encrypted embeddings	Secure but slow
Sharma et al. [35]	Lightweight CNN model	Decent accuracy; lacks encryption
Aliradi et al. [36]	LGBQ: Gabor + LBP + LPQ	Feature fusion; lacks privacy protection
Chen et al. [37]	HEFDIVS with FaceNet + ISOMAP + CKKS FHE	Dual encryption

pose variations. This enhances the accuracy and reliability of facial analysis, even under challenging conditions, and contributes to the development of more resilient FR models.

Ke et al. [28] proposed LocalFace. This model focuses on key facial regions. It works well even with occlusions like masks or sunglasses. Attention modules help improve accuracy. Huang et al. [29] solved the low-light problem in FR. They introduced a lightweight feature restoration network. It improves facial details before recognition. Their method does not need large datasets or heavy computation. Ali et al. [21] used SVM with SqueezeNet. They reported 93.39% accuracy on the 105PinsFace dataset. Miakshyn et al. [22] used FaceNet and OpenFace. They achieved 83.53% accuracy on the 105PinsFace dataset. Yang et al. [33] proposed HeadPose-Softmax. It is a loss function based on curriculum learning. It helps the model learn better from difficult poses. It also improves multi-view face detection. Sharma et al. [35] used a lightweight CNN. This method reached 89.38% accuracy on the 105PinsFace dataset. CNNs now play a big role in FR. They extract complex features and improve performance.

Cheema et al. [31] proposed the DNDR network. It helps detect disguised faces in different image types. Their method uses deep feature differences, joint loss, and multi-stage training. This improves both speed and accuracy. Alirezazadeh et al. [25] and Albraikan et al. [26] proposed methods for masked FR and emotion detection. They used ACML, DenseNet121, and ResNet152v2. These models showed strong results. However, validation on large datasets remains limited. FR in cloud systems raises privacy issues. FHE helps protect data. However, it is often slow. Many FR systems prioritise accuracy over privacy. This puts user data at risk. Privacy-preserving FR can fix this. It keeps the data safe at every stage using encryption. Mi et al. [27, 34] used FHE with DL-based FR. They enabled the encrypted computation of facial features. Boddeti et al. [17] improved performance using batching and dimensionality reduction. This made FHE more practical for real-world use.

Ma et al. [15] demonstrated the success of combining homomorphic encryption (HE) and deep learning (DL). But more work is needed to check scalability and usability. Xie et al. [32] proposed a framework for edge-based FR. It used local differential privacy to protect face images and training models. Khan et al. [23] secured the feature extraction process. They used an encrypted system based on Local Ternary Pattern (LTP). Chen et al. [37] proposed HEFDIVS, which combines FaceNet, ISOMAP, and CKKS-based FHE with hybrid encryption. It achieves fast, accurate verification (95–97%), but the dual encryption and additional processing add complexity, making it less suitable for resource-limited devices. Hu et al. [39] proposed a privacy-preserving face recognition system that supports different models like MobileFaceNet, ResNet-18, and ResNet-50. It uses a simple encryption method to protect facial features. Wang et al. [40] introduced a privacy-preserving face generation method for robust authentication by creating synthetic faces that hide identity information. However, balancing privacy and accuracy requires high computational power.

Face recognition systems have made significant progress in terms of accuracy; however, many still lack robust privacy and security measures. In most existing methods, face images or features are processed without encryption, which exposes sensitive user data to risk, particularly in cloud-based applications. While FHE enables recognition of encrypted data without revealing it, it is often slow and requires significant computation. This makes it challenging to use in real-time systems or on devices with limited

resources. Some newer methods try to combine deep learning with encryption, but these are often too complex or not scalable for practical use. In addition, only a few systems utilise advanced deep models, such as FaceNet or ArcFace, within a fully secure and privacy-preserving framework. Due to these challenges, there is a clear need for face recognition methods that protect user data through efficient encryption while maintaining high recognition performance.

Girdhar et al. [41] provide an overview of recent progress in biomedical trait-based human identification, showing how technologies such as machine learning and advanced sensors have improved biometric systems used in security and authentication. Chowdhury et al. [42] focus on the challenge of protecting images stored in the cloud and introduce the PwLMμ-TPE encryption method, which keeps image thumbnails visible for easy use while still protecting the full image through strong encryption. Dhar et al. [43] review both digital and quantum watermarking techniques and explain how quantum approaches, based on principles such as superposition and entanglement, can provide stronger protection against tampering than traditional methods. Together, these studies highlight ongoing efforts to improve security, privacy, and data protection across different digital applications.

2.1 Notation

Table 2 refers to the notations used in this system

3 Privacy-preserving face recognition via HE and deep learning

The proposed system employs a client–server setup, as illustrated in Fig. 1, which provides an overview of the enrollment and verification stages. The system ensures secure face authentication using FHE while maintaining user privacy. The system has three main components: the client device, the Trusted Authenticator (TA), and the Cloud Server. During enrolment, the client device captures the user's face image and extracts facial features using a Deep Convolutional Neural Network (DCNN), such as FaceNet or ArcFace. These models are chosen for their proven ability to produce compact, highly discriminative embeddings. These extracted features are then sent to the TA, who applies LPP to reduce dimensionality, making the features more efficient for storage and comparison. The reduced features, known as RLpp, are encrypted using FHE and stored on the Cloud Server as E(R) along with the user's ID.

Table 2 Notations used in the proposed framework

Notation	Description
P_{lpp}	Probe template stored
R_{lpp}	Reference template
P_k	Public key
S_k	Secret key
P_i	Plaintext
E	Encryption
D	Decryption
D_t	Distance between templates
T	Matching threshold
$E(P)$	Encrypted probe template
$E(R)$	Encrypted reference template
N	Polynomial modulus degree
s	Coefficient modulus bits

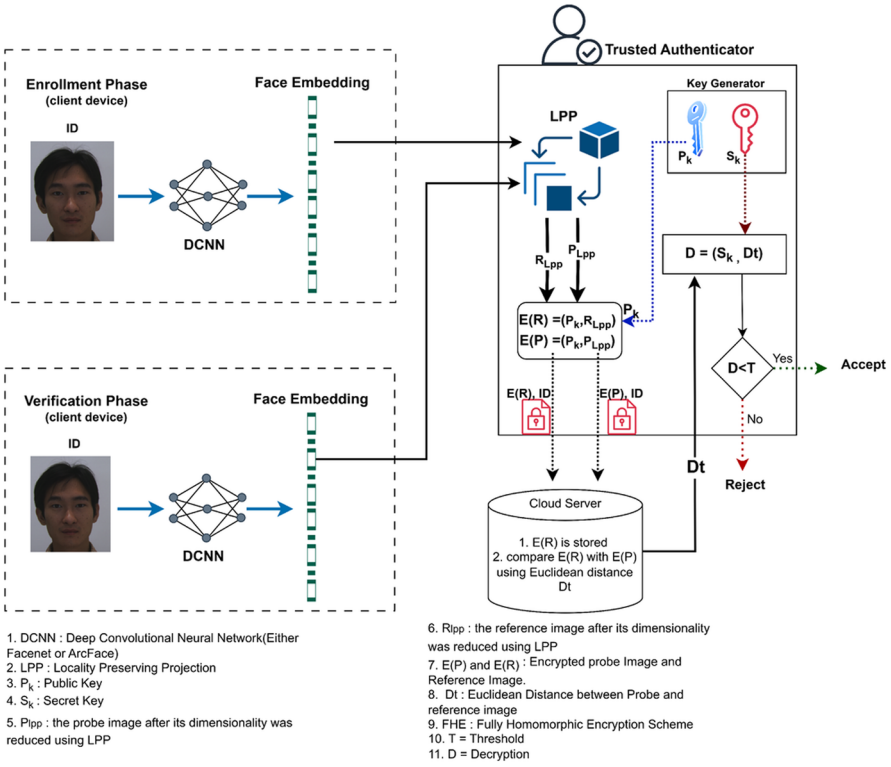


Fig. 1 Overview of the enrollment and verification phases in the proposed face authentication system

During the verification phase, the user provides a new face image, which is then processed on the client device using the same feature extraction process. The extracted feature vector is sent to the TA, where LPP is applied to obtain PLpp, which is then encrypted to form E(P). This encrypted probe embedding is sent to the cloud server for matching. The Cloud Server retrieves the corresponding E(R) from storage and computes the Euclidean distance between E(R) and E(P). The encrypted distance value, Dt, is forwarded to the TA, which decrypts it using a secret key. Finally, the system compares the decrypted value (D) with a predefined threshold (T) to determine whether authentication is successful. If D is below the threshold, the user is accepted; otherwise, authentication is denied. This architecture ensures that face data remains protected throughout the process, preventing unauthorised access while maintaining efficient and accurate authentication.

3.1 Assumptions

In the proposed approach, the following are the assumptions:

- The client device is considered fully trusted during both the enrollment and authentication phases.
- The keys are securely stored in the trusted authenticator's local memory throughout enrollment and authentication.
- The cloud server follows an honest-but-curious model, meaning it performs the required computations-such as calculating the Euclidean distance-accurately but may attempt to learn information from the data it processes. However, the computed distances remain correct and reliable.

- The keys are generated and stored inside a trusted authenticator. The client never keeps keys.
- All communications between the client, the trusted authenticator, and the cloud server occur over secure, trusted channels. As part of the authentication process, the client device transmits feature embeddings to the trusted authenticator via this secure connection.

3.2 Preparation of face images for feature extraction

Before extracting facial features using deep learning models, all input images are preprocessed to ensure consistency and improve recognition performance. Preprocessing helps the models focus on the actual facial characteristics by reducing variations caused by background, lighting, and pose. The first step is face detection, where the facial region is identified and cropped from the original image. This ensures that only the relevant part of the image is used for analysis. Following this, face alignment is performed using key facial landmarks such as the eyes and nose. Alignment standardises the face's orientation, reducing the impact of head tilts or rotations.

Once aligned, the images are resized to match the input size expected by the models. For example, FaceNet requires images of size 160×160 pixels, while ArcFace typically uses 112×112 pixels. The final step is normalisation, where pixel values are scaled to the range of -1 to 1 to minimise the impact of lighting variations and ensure consistent model performance across images. These preprocessing steps, illustrated in Fig. 2, are essential for improving the quality of the extracted features and ensuring that face recognition models perform effectively.

3.3 Generation of facial embedding vectors

The proposed system leverages two well-established deep learning models, FaceNet [44] and ArcFace [45], to extract robust and discriminative facial features for face recognition (FR) tasks. These models are selected due to their performance on large-scale face verification benchmarks and their learning strategies.

FaceNet [17, 44] is based on the InceptionResNetV1 architecture and was pre-trained on the VGGFace2 dataset. While the original FaceNet architecture was designed to output 128-dimensional embeddings for efficient processing, the FaceNet model used in the proposed system produces 512-dimensional embeddings. This expanded embedding size enables the model to capture more fine-grained facial details, thereby improving recognition accuracy, especially when working with large, diverse datasets. The model employs triplet loss to ensure that embeddings from the same identity are closer together, while embeddings from different identities are pushed apart. Triplet-based

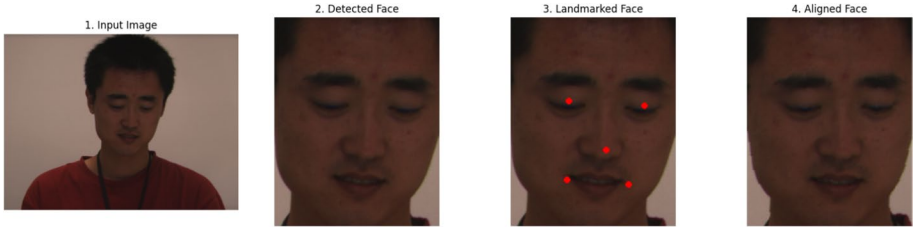


Fig. 2 Illustration of the face preprocessing pipeline using FaceNet on the CASIA-3D Face dataset. The process includes face detection, alignment, resizing, and feature embedding extraction

learning focuses on preserving global distance relationships in Euclidean space. All extracted features are L2-normalised, making them suitable for Euclidean-based similarity comparisons.

Unlike existing FaceNet-based privacy-preserving approaches that operate directly on raw or PCA-reduced embeddings, the proposed system introduces Locality Preserving Projection (LPP) on deep embeddings before encryption. This step preserves local manifold structure while significantly reducing dimensionality, enabling compact encrypted templates and improving the efficiency of CKKS-based matching. The integration of FaceNet with LPP and optimised CKKS encryption has not been explored in prior work, representing a key novelty of the proposed framework.

ArcFace is integrated through the FaceAnalysis module from the InsightFace framework. It is built on a ResNet backbone and trained using the MS-Celeb-1 M dataset. ArcFace outputs 512-dimensional embeddings by default and applies an additive angular margin loss to enhance intra-class compactness and inter-class separability in the embedding space, thereby enhancing class separation on a hypersphere. This loss formulation yields highly discriminative embeddings that perform well under challenging conditions, such as variations in pose, illumination, and facial expression. Employing both models provides flexibility in embedding strategies and enables comparative performance analysis across datasets and conditions.

ArcFace is used in the proposed system due to its additive angular-margin formulation produces highly discriminative and well-separated embeddings, and its implementation within the InsightFace framework [46–48] provides stable performance on both balanced and imbalanced datasets. This robustness arises from ArcFace's feature and weight normalisation on the hypersphere, combined with a consistent angular margin that reduces bias toward classes with more samples and maintains uniform decision boundaries across identities. Earlier margin-based approaches, such as SphereFace [49] and CosFace [50], also contributed to improving face recognition, but each presents limitations: SphereFace uses a multiplicative angular margin that can lead to optimisation instability, while CosFace applies a cosine margin in the similarity space rather than directly in the angular domain, resulting in less consistent geometric separation. ArcFace addresses these issues by enforcing a clean and geometrically intuitive additive angular margin, enabling more stable optimisation, stronger inter-class separability, and reliable embedding quality even under dimensionality reduction and encrypted matching.

3.4 Projection of high-dimensional embeddings

LPP is particularly effective for dimensionality reduction in FR systems, as it preserves local neighbourhood structures while reducing the dimensionality of high-dimensional feature embeddings from models such as FaceNet and ArcFace. In comparison, Principal Component Analysis (PCA) [51] captures global variance but may lose fine-grained local details, Linear Discriminant Analysis (LDA) [52] requires class labels and prioritises inter-class separability, and Independent Component Analysis (ICA) [53] assumes statistical independence among features. Unlike these techniques, LPP [54] retains geometric locality and the intrinsic manifold structure of the data, ensuring a discriminative and compact feature space for robust FR. By preserving intrinsic relationships within the data, LPP effectively handles variations in pose, lighting, and expression, enhancing computational efficiency while maintaining the manifold structure of facial images for

accurate analysis. The procedure for reducing the dimensionality of feature embeddings is described in Algorithm 1.

Require: $X \in \mathbb{R}^{n \times d}$, ϵ, k.
Ensure: $Y \in \mathbb{R}^{n \times k}$.
Construct Adjacency Graph:
Compute distance matrix $D_{ij} \leftarrow \|x_i - x_j\|_2$.
$W_{ij} \leftarrow 1$ if $D_{ij} \leq \epsilon$, else 0.
Set $W_{ii} \leftarrow 0$.
Compute Weight Matrix:
if using heat kernel **then**
  $W_{ij} \leftarrow \exp\left(-D_{ij}^2/t\right)$.
else
  $W_{ij} \leftarrow 1$ if connected, 0 otherwise.
end if
Compute Graph Laplacian:
$D_{ii} \leftarrow \sum_j W_{ij}$.
$L \leftarrow D - W$.
Solve Generalized Eigenvalue Problem:
Solve $X^\top L X a = \lambda X^\top D X a$.
Select k smallest eigenvectors.
Compute Projection:
$A \leftarrow$ selected eigenvectors.
$Y \leftarrow XA$.
Return Y.

Algorithm 1 Locality preserving projection (LPP)

In the proposed system, an ϵ-neighbourhood approach is used to define local relationships among data points. Data points are treated as neighbours if the Euclidean distance between them is less than or equal to $\epsilon = 0.5$. This threshold was empirically chosen to balance graph sparsity and connectivity, allowing only locally relevant data to influence the projection matrix. To quantify similarity between neighbours, the heat kernel weighting function is utilised:

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{t}\right) \tag{1}$$

where t is a scaling parameter that governs the smoothness of the weight decay. The use of this continuous kernel ensures that closer points have a stronger influence on the local graph structure, enhancing the algorithm's sensitivity to fine local variations in facial features.

Following the construction of the weight matrix and graph Laplacian, LPP solves a generalised eigenvalue problem of the form:

$$X^\top L X a = \lambda X^\top D X a \tag{2}$$

Here, $L = D - W$ is the graph Laplacian, and D is the diagonal matrix containing the sum of weights for each node. The solution yields a set of eigenvectors corresponding to the smallest eigenvalues, which are used to construct a projection matrix. This matrix is then applied to map the original high-dimensional deep features into a lower-dimensional space.

In our experiments, the 512-dimensional deep features generated by FaceNet and ArcFace are reduced to 256 dimensions. This dimensionality was selected based on extensive empirical evaluation, offering a trade-off between recognition performance and computational efficiency. The resulting compressed vectors reduce storage and processing overhead. By preserving local geometric structures, LPP ensures that the most informative aspects of facial features are retained, even after significant dimensionality reduction, as depicted in Fig. 3.

3.4.1 Rationale for using LPP

Instead of Direct 256-dimensional Embeddings, FaceNet and ArcFace are trained on large-scale face datasets and naturally output 512-dimensional embeddings. This representation is well validated for generalisation across different datasets. Modifying these deep models to produce 256-dimensional vectors would require extensive retraining on millions of images, which is computationally expensive and may reduce generalisation performance. LPP provides a training-free and structure-preserving dimensionality reduction method. It reduces the embeddings to 256 dimensions while maintaining the local geometric relationships between samples. This is important for face recognition because the discriminative structure of the feature space must be preserved even after encryption. Reducing the feature dimension also has practical benefits in the encrypted

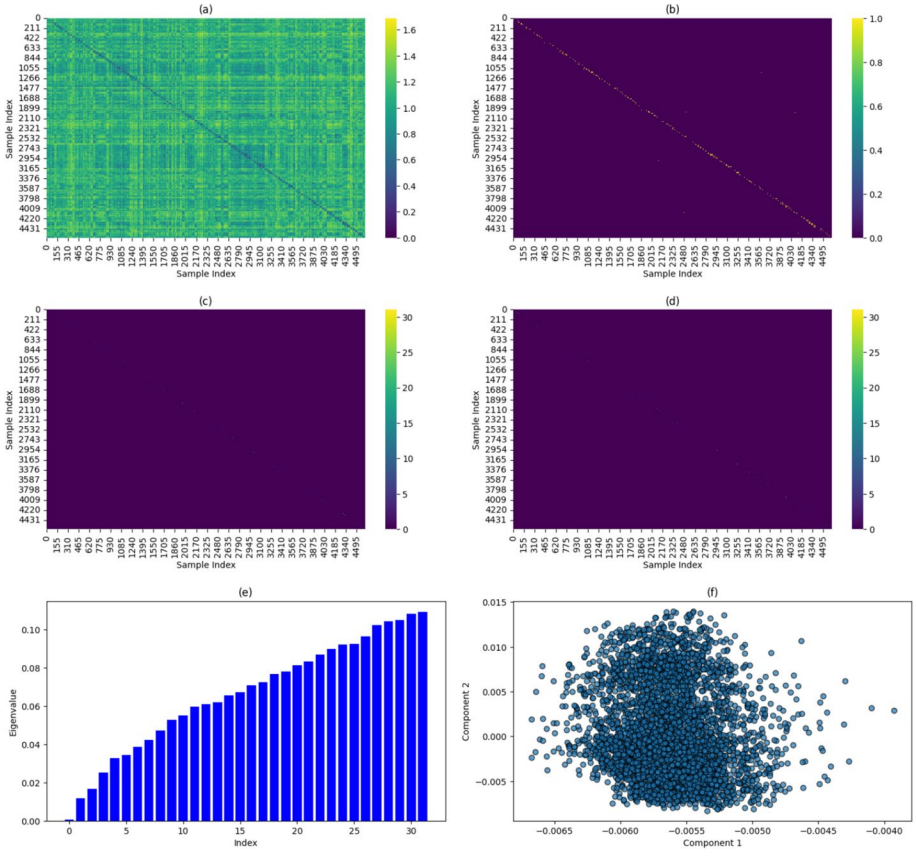


Fig. 3 Visualisation of the locality preserving projection (LPP) pipeline. The figure illustrates the transformation of the input feature matrix into a lower-dimensional space while preserving local neighbourhood structures. Subfigures show the step-by-step process: **a** Distance matrix capturing pairwise similarities, **b** Weight matrix representing local affinities, **c** Degree matrix showing node connectivity, **d** Graph Laplacian encoding the geometric structure, **e** Eigenvalues from decomposition, and **f** Final projected data Y in the reduced space

domain. Encrypting a 256-dimensional vector requires less memory and computation than encrypting a 512-dimensional vector. This leads to smaller ciphertext sizes and faster encrypted distance computation. Therefore, using 512-D pretrained embeddings followed by LPP offers a good balance between accuracy, privacy, and efficiency.

3.5 Homomorphic encryption of facial features

FHE ensures that sensitive facial information stays secure and private throughout the process. Even if someone accesses the data during processing, they cannot understand it because it remains encrypted. After dimensionality reduction using LPP, complex numbers are often encountered, which can be efficiently handled using CKKS FHE. This capability makes CKKS FHE particularly suitable for privacy-preserving FR systems, as it supports operations on encrypted data, including computations involving complex numbers. Its ability to perform encrypted computations without decryption ensures end-to-end security, making it an ideal choice for systems that require both high-level privacy and robust computational capabilities. By addressing the challenges posed by complex numbers and ensuring the secure handling of sensitive information, FHE significantly enhances the reliability and security of FR systems.

3.5.1 FHE

FHE schemes enable computation on encrypted data without decryption. Several schemes exist, each with strengths and limitations. The Brakerski–Gentry–Vaikuntanathan (BGV) scheme [55] and its variants like BFV (Fan–Vercauteren, an FHE scheme, a variant of BGV) excel at integer arithmetic but contend with noise growth. Similarly, the Fan–Vercauteren (FV) scheme [56] faces limitations due to noise growth. The CKKS scheme [57] specialises in approximate arithmetic on real and complex numbers, making it suitable for applications like machine learning and signal processing, and particularly advantageous for tasks involving complex-valued data like FR and medical image analysis. While efficient for approximate computations, CKKS, like the closely related HEAAN scheme [58]. The TFHE scheme [59] is optimised for binary operations, excelling in logic-based computations but potentially limited in applicability to other data types. Among these, the CKKS scheme stands out for its ability to handle approximate arithmetic on real and complex numbers. This makes CKKS particularly well-suited for modern applications, such as privacy-preserving machine learning, where fixed-point operations are essential. CKKS introduces the concept of packed computations, enabling multiple plaintext values to be processed in parallel, which significantly enhances efficiency.

3.5.2 CKKS-based secure embedding and distance computation

The CKKS HE scheme allows approximate arithmetic operations to be performed directly on encrypted real-valued vectors.

Let $e_1, e_2 \in \mathbb{R}^d$ be two facial embeddings extracted from deep learning models. Each embedding is encrypted using the CKKS scheme to produce corresponding ciphertexts C_1 and C_2 as follows:

$$C_i = \text{Encrypt}(e_i, P_k), \quad i \in \{1, 2\} \tag{3}$$

where P_k denotes the public key used for encryption.

To compute the similarity between these encrypted embeddings, the squared Euclidean distance is approximated homomorphically. First, the difference between the encrypted vectors is computed:

$$\Delta C = C_1 - C_2 \tag{4}$$

Then, the element-wise square is performed in the encrypted domain:

$$D_{\text{enc}} = \Delta C \cdot \Delta C \tag{5}$$

Since the ciphertext size grows after multiplication, a relinearization operation is applied to maintain computational efficiency:

$$D_{\text{enc}} \leftarrow \text{Relinearize}(D_{\text{enc}}) \tag{6}$$

The result D_{enc} is a ciphertext containing the squared element-wise differences between the original embeddings. This encrypted output is then decrypted and decoded using the secret key S_k:

$$\tilde{D} = \text{Decrypt}(D_{\text{enc}}, S_k) \tag{7}$$

$$D = \text{Decode}(\tilde{D}) \tag{8}$$

The decrypted vector $D \in \mathbb{R}^d$ approximates the squared differences of the original embeddings. The Euclidean distance between e_1 and e_2 is then computed as:

$$\text{Distance}(e_1, e_2) \approx \sqrt{\sum_{j=1}^{d} D_j} \tag{9}$$

This approach ensures that facial templates remain encrypted throughout the matching process, significantly enhancing the privacy and security of the proposed system. By leveraging the properties of CKKS, secure and efficient similarity comparisons can be performed without ever exposing sensitive data.

3.5.3 Justification and rationale

The CKKS homomorphic encryption scheme is used because it supports approximate computations over encrypted real numbers, making it suitable for encrypting facial feature data. The polynomial modulus degree $n = 8192$ strikes a balance between strong security and reasonable computation time. The modulus chain bits $[60, 40, 40, 60]$ provide sufficient space to perform multiple encrypted operations, such as addition and multiplication, while keeping noise levels low. The scale factor $s = 2^{40}$ helps keep precision when encoding decimal numbers, flattening the feature vectors before encryption allows efficient vector operations with CKKS. After multiplication, a relinearization step reduces ciphertext size and noise, improving the accuracy and speed of encrypted distance calculations. Finally, decryption and decoding with the secret key recover the approximate distance value in plaintext. These choices enable the secure computation of face recognition distances on encrypted data, without exposing sensitive face information.

Table 3 CKKS parameter tuning and its effect on security and performance

N	Coeff mod bit sizes	Security level (bits)	Enc. (s)
4096	[60, 40, 40, 60]	~ 80	0.016
4096	[40, 21, 21, 21, 21, 40]	~ 80	0.032
4096	[40, 20, 20, 40]	~ 80	0.033
8192	[60, 40, 40, 60]	~ 128	0.046
8192	[40, 21, 21, 21, 21, 40]	~ 128	0.059
8192	[40, 20, 20, 40]	~ 128	0.065
16,384	[60, 40, 40, 60]	~ 192	0.082
16,384	[40, 21, 21, 21, 21, 40]	~ 192	0.097
16,384	[40, 20, 20, 40]	~ 192	0.099

Table 4 Comparison of CKKS scaling factors across different aspects

Scaling factor	Precision	Noise growth	Rescaling stability	Performance
2^{21}	Low	High	Less stable	Fast
2^{40}	High	Low	Stable	Slower

Several parameter choices for the CKKS scheme were assessed to analyse their performance, accuracy, and computational costs, as depicted in the Table 3. In particular, the three levels of modulus bit sizes tested, 4096, 8192, and 16,384, three different types of configurations were analysed: [60, 40, 40, 60], [40, 21, 21, 21, 21, 40], and [40, 20, 20, 40]. The lower degree configuration (4096) had the fastest computation times and the smallest ciphertext size, but its security was about 80 bits. Increasing the polynomial modulus degree to 8192 achieved a more balanced performance/precision trade-off, providing ($\sim$ 128 bit) security with a moderate resource footprint. The 16,384 offered the best security level ($\sim$ 192 bits), more precision, but a larger performance overhead and ciphertext sizes. With respect to coefficient modulus bit-size variations, configurations that were deeper at the bit level, such as [40, 21, 21, 21, 21, 40] and [40, 20, 20, 40], offered better numerical accuracy at the cost of more latency. This tuning process highlights the need for parameters that are reasonable in terms of precision, security, and performance.

The effect of scaling factor choice in the CKKS scheme was studied here by testing both 2^{21} and 2^{40} configurations using various polynomial modulus degrees and coefficient modulus chains, which is depicted in the Table 4. It was found that using a smaller scaling factor (2^{21}) resulted in faster execution and smaller ciphertext sizes. Still, it also led to more noise growth and less consistent rescaling during homomorphic operations. On the other hand, a higher scaling factor of 2^{40} provided better numerical precision and stable rescaling, which are essential for deeper computations on encrypted data. The higher scaling factor was more expensive computationally, but the greater accuracy and robustness made it a more logical choice. Thus, all subsequent assessments in the encrypted space used 2^{40} for the sake of reliability and security.

4 Experimental setup and result analysis

4.1 Datasets

4.1.1 CASIA 3D face dataset

The CASIA 3D Face dataset was collected from the Centre for Biometrics and Security Research (CBSR) [60]. It includes 4624 scans of 123 subjects taken with a 3D digitiser. The dataset encompasses variations in poses, expressions, and lighting conditions,

combining these factors. People wearing glasses had additional samples taken. Each person's dataset has 37 or 38 samples.

4.1.2 105Pins face dataset

The 105Pinsface Dataset was sourced from Kaggle [61]. This dataset comprises 17,534 facial images extracted from Pinterest, featuring 105 celebrities. The diverse origins of these images provide both opportunities and challenges for FR tasks.

4.1.3 Faces94 dataset

The Faces94 dataset, a facial image collection designed for evaluating face recognition models under controlled conditions, was downloaded from http://cswww.essex.ac.uk/mv/allfaces/faces94.zip or https://github.com/chenyijia1997/Face-datasets. It contains images of 153 individuals, comprising 133 males and 20 females, with 20 images per person. Each image has a fixed resolution of 180 $\times$ 200 pixels and features a uniform green background to minimise background distractions. Although the images are captured in a controlled setting, they include natural variations in facial expressions and slight changes in head pose. These subtle differences make the dataset suitable for studying intra-class variability.

4.1.4 Labelled faces in the wild (LFW)

The LFW dataset [62] is downloaded from Kaggle, and it is commonly used to test face recognition algorithms in challenging, real-world settings. Developed by the University of Massachusetts, Amherst, it includes 13,233 face images representing 5749 individuals. These images were gathered from the internet and initially processed using the Viola–Jones face detection algorithm to ensure proper alignment and cropping. Although the dataset contains many individuals with only one image, a subset of 1680 individuals has at least two images, making it suitable for verification tasks. Among the available versions, the deep-funnelled version provides better facial alignment and has been shown to improve recognition accuracy. For this study, we use a filtered subset of the deep-funnelled images, consisting of 779 classes, each containing exactly two samples, to evaluate the performance of the proposed privacy-preserving face recognition system.

4.2 System configuration

All experiments in the proposed approach were conducted on a Dell Precision 3660 workstation running Windows 10 Pro (64-bit). The system was equipped with an Intel Core i7-12700 (12th generation) processor and 32 GB of RAM. It consists of an NVIDIA GeForce RTX 3060 GPU with 12 GB of dedicated VRAM and an Intel UHD Graphics 770 integrated GPU. The software environment was set up using Python 8. The required deep learning tools were installed using PyTorch (version 2.2.2) and Torchvision (version 0.17.2). The tenseal library with CKKS enabled was used for all homomorphic encryption operations. This configuration formed the computational environment used throughout all experimental stages of the proposed system.

4.3 Performance metrics

The performance of proposed system is evaluated using Eqs. (10)–(15),

False acceptance rate (FAR): The percentage of legitimate users (genuine) who are mistakenly denied access by the system.

$$\text{FAR} = \frac{\text{No. of impostor scores above the T}}{\text{Total no. of impostor scores}}. \tag{10}$$

False rejection rate (FRR): The percentage of legitimate users (genuine) who are mistakenly denied access by the system.

$$\text{FRR} = \frac{\text{No. of genuine scores below the T}}{\text{Total no. of genuine scores}}. \tag{11}$$

Equal error rate (EER): The rate at which both FAR and FRR are the same. A lower EER signifies better accuracy of the biometric system.

Accuracy: The percentage of correctly classified instances (both genuine and impostors) in the system.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{12}$$

Precision: The proportion of correctly identified genuine instances out of all instances classified as genuine.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{13}$$

Recall (sensitivity): The proportion of correctly identified genuine instances out of all actual genuine instances.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{14}$$

F1 score: The harmonic mean of Precision and Recall.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{15}$$

4.4 Results comparison

The proposed system was evaluated using four benchmark datasets-CASIA 3D Face, 105 Pins Face, Faces94, and LFW-to test its accuracy, privacy preservation, and computational efficiency. Across all datasets, ArcFace consistently outperformed FaceNet, especially at LPP dimension 256, making it the most balanced configuration in terms of performance and efficiency.

The CASIA 3D Face dataset contains depth and pose variations that challenge the robustness of facial features. As reported in Table 5, ArcFace with 256-D achieved a notably low EER of 0.0077%, reflecting strong decision boundary placement between genuine and impostor pairs in the projected subspace. This dimension also delivered a high recall of 99.17%, showing the system's ability to correctly identify genuine matches despite 3D distortions. FaceNet also benefited from the 256-D projection, achieving its best accuracy (98.67%) and improved class separation due to the preserved geometric structure provided by LPP. These results show that ArcFace effectively captures complex

Table 5 Evaluation metrics of FaceNet and ArcFace on CASIA 3D face dataset

Model	LPP-D	EER %	Accuracy %	Precision %	Recall %	F1-score %	Enc. time (s)	Comp. time (s)
FaceNet	32	0.040	96.52	83.21	95.93	88.12	0.2071	0.3368
	64	0.040	96.47	82.14	95.12	88.09	0.4142	0.6736
	128	0.040	96.79	84.12	95.12	89.23	0.8285	1.3473
	256	0.040	98.67	89.25	95.12	92.08	1.6570	2.6947
	512	0.040	98.51	90.87	95.12	92.94	3.3141	5.3895
ArcFace	32	0.032	96.73	85.22	96.69	90.52	0.2071	0.3368
	64	0.016	98.36	91.13	98.35	94.60	0.4142	0.6736
	128	0.0083	99.16	96.20	99.17	98.25	0.8285	1.3473
	256	0.0077	99.30	97.35	99.17	97.66	1.6570	2.6947
	512	0.0082	99.02	96.80	99.17	97.97	3.3141	5.3895

Enc. time: Encryption time in seconds

Comp. time: Distance computation time in seconds

Table 6 Evaluation metrics of FaceNet and ArcFace on 105PinsFace dataset

Model	LPP-D	EER %	Accuracy %	Precision %	Recall %	F1-score %	Enc. time (s)	Comp. time (s)
FaceNet	32	0.110	91.47	66.18	85.71	74.73	0.2071	0.3368
	64	0.080	93.29	71.36	89.00	79.33	0.4142	0.6736
	128	0.060	93.54	74.50	93.30	82.91	0.8285	1.3473
	256	0.045	95.60	79.46	94.29	85.21	1.6570	2.6947
	512	0.050	95.60	77.49	94.29	86.27	3.3141	5.3895
ArcFace	32	0.090	90.63	68.02	90.48	77.60	0.2071	0.3368
	64	0.070	93.00	72.65	92.38	81.37	0.4142	0.6736
	128	0.020	97.20	84.68	97.14	90.51	0.8285	1.3473
	256	0.015	98.06	90.17	98.10	93.99	1.6570	2.6947
	512	0.015	98.06	89.43	98.10	93.56	3.3141	5.3895

Enc. time: Encryption time in seconds

Comp. time: Distance computation time in seconds

spatial features and maintains high performance even after dimensionality reduction and encryption.

The 105PinsFace dataset includes variations in illumination and expression across constrained identities. As shown in Table 6, ArcFace achieved its best performance at 256 dimensions with an EER of 0.010%, an accuracy of 98.06%, and an F1-score of 93.99%. This demonstrates its strong ability to distinguish genuine users from impostors in moderately unconstrained conditions. The results suggest that LPP at 256-D effectively preserves the discriminative structure of the data by maintaining intra-class compactness and inter-class separation. FaceNet also reached its peak performance at this dimension, confirming that 256-D provides an optimal balance between dimensionality reduction and feature preservation for both models.

Faces94, a relatively clean and controlled dataset, offers limited variation, making it easier for models to achieve high performance across dimensions. However, at 256-D, FaceNet attained its highest accuracy of 99.61%, with consistently strong recall and F1-score of 98.71%. The evaluation results can be found in Table 7. However, ArcFace at 256 dimensions achieved an accuracy of 99.82% with minimal EER 0.0059%. The precision and F1-score values confirmed consistent, reliable performance. This demonstrates that even in less challenging environments, ArcFace maintains superior feature representation and matching consistency.

Table 7 Evaluation metrics of FaceNet and ArcFace on Faces94 dataset

Model	LPP-D	EER %	Accuracy %	Precision %	Recall %	F1-score %	Enc. time (s)	Comp. time (s)
FaceNet	32	0.0064	99.39	91.03	99.35	94.99	0.2071	0.3368
	64	0.0058	99.44	92.71	99.41	95.95	0.4142	0.6736
	128	0.0059	99.46	93.22	99.41	96.21	0.8285	1.3473
	256	0.0059	99.61	98.03	99.41	98.71	1.6570	2.6947
	512	0.0059	99.61	94.48	99.41	96.89	3.3141	5.3895
ArcFace	32	0.00583	99.42	92.07	99.42	95.59	0.2071	0.3368
	64	0.00582	99.43	92.31	99.41	95.72	0.4142	0.6736
	128	0.00581	99.43	92.45	99.41	95.81	0.8285	1.3473
	256	0.00590	99.82	93.75	99.41	96.51	1.6570	2.6947
	512	0.00590	99.82	91.03	99.41	94.99	3.3141	5.3895

Enc. time: Encryption time in seconds

Comp. time: Distance computation time in seconds

Table 8 Evaluation metrics of FaceNet and ArcFace on LFW dataset

Model	LPP-D	EER %	Accuracy %	Precision %	Recall %	F1-score %	Enc. time (s)	Comp. time (s)
FaceNet	32	0.0840	91.61	86.92	91.53	89.16	0.2071	0.3368
	64	0.0720	92.89	89.13	92.68	90.87	0.4142	0.6736
	128	0.0680	93.33	89.94	93.07	91.48	0.8285	1.3473
	256	0.0652	95.72	92.49	95.12	93.79	1.6570	2.6947
	512	0.0652	95.72	91.91	93.32	92.61	3.3141	5.3895
ArcFace	32	0.0730	94.20	87.62	91.78	89.66	0.2071	0.3368
	64	0.0440	95.60	89.33	95.51	92.32	0.4142	0.6736
	128	0.0380	96.79	91.34	96.15	93.68	0.8285	1.3473
	256	0.0344	97.07	92.86	96.53	94.66	1.6570	2.6947
	512	0.0344	97.10	91.27	96.53	93.83	3.3141	5.3895

Enc. time: Encryption time in seconds

Comp. time: Distance computation time in seconds

The LFW dataset presents real-world challenges, including variations in lighting, pose, and expression. As shown in Table 8, ArcFace achieved strong performance at 256 dimensions, with an accuracy of 97.07%, F1-score of 94.66%, and the lowest EER of 0.034%, demonstrating its ability to generalise well under unconstrained conditions. FaceNet also performed best at this dimensionality, reaching an accuracy of 95.72%.

ArcFace with 256-D features consistently gave the best results across all datasets. It preserved identity-specific information even after dimension reduction and data encryption. This setup supports fast, secure, and accurate face recognition, making it ideal for real-time and large-scale systems. The 256-dimensional features preserved important local patterns in the data, helping the system match similar faces more accurately. Unlike lower dimensions that lose useful details or higher ones that add extra noise, 256-D offered the best balance, keeping the system fast, accurate, and privacy-friendly.

The performance of the proposed system has been compared with several existing methods, as illustrated in Table 9. The results clearly show the strength of the proposed models—Facenet and ArcFace—in achieving high verification rates. On the CASIA-3D dataset, earlier methods, such as multi-scale FPFH with template matching and handcrafted features such as LGBQ, Gabor, LBP, and LPQ, have achieved good performance, with the highest verification rate of 99.23% reported by Aliradi et al. However, the proposed ArcFace model achieved 99.30%, slightly outperforming the existing best. This shows that DL-based methods, especially ArcFace, are more effective for 3D FR.

Table 9 Comparison with existing methods

Dataset	Authors	Model	Acc.	Prec.	Recall	F1
CASIA-3D	Chen et al. [30]	CNN (2D → 3D)	96.75	–	-	–
	Aliradi et al. [36]	LGBQ + Gabor, LBP, LPQ	99.23	–	-	–
	Proposed	FaceNet	98.67	89.25	95.12	92.08
	Proposed	ArcFace	99.30	97.35	99.17	97.66
105PinsFace	Ali et al. [21]	SVM + Inception/SqueezeNet	93.08, 93.39	–	-	–
	Miakshyn et al. [22]	FaceNet + OpenFace	85.53	90.67	58.12	70.84
	Sharma et al. [35]	Lightweight CNN	89.38	–	-	–
	Najeeb et al. [63]	CNN	91.29	–	-	–
	Proposed	FaceNet	95.60	79.46	94.29	85.21
	Proposed	ArcFace	98.06	90.17	98.10	93.99
Faces94	Yang et al. [64]	FaceNet + CKKS	96.74	–	-	–
	Hu et al. [39]	MobileFaceNet/ResNet	96.15	–	-	–
	Chen et al. [37]	FaceNet + Hybrid Enc.	96.98	–	-	–
	Proposed	FaceNet	99.61	98.03	99.41	98.71
	Proposed	ArcFace	99.82	93.75	99.41	96.51
LFW	Miakshyn et al. [22]	FaceNet + OpenFace	91.73	85.71	87.18	86.44
	Yang et al. [64]	FaceNet + CKKS	96.67	–	-	–
	Hu et al. [39]	MobileFaceNet/ResNet	95.50	–	-	–
	Chen et al. [37]	FaceNet + Hybrid Enc.	95.45	–	-	–
	Proposed	FaceNet	95.72	92.49	95.12	93.79
	Proposed	ArcFace	97.07	92.86	96.53	94.66

For the 105PinsFace dataset, which has greater variation and is considered more challenging, traditional models achieved lower verification rates of 83.53–93.39%. The proposed models showed clear improvements, with Facenet achieving 95.60% and ArcFace achieving 98.06%. This shows that the proposed system is capable of handling complex and diverse face data more effectively. Overall, the results across all three datasets demonstrate that the proposed models-especially ArcFace-consistently outperform existing methods. This confirms the effectiveness and generalisation ability of the proposed approach for FR tasks.

Compared with existing approaches, the proposed system demonstrated a notable improvement on the Faces94 dataset, which comprises frontal face images with minimal variation. Existing methods that used FaceNet with CKKS encryption or hybrid approaches achieved approximately 96–97% accuracy; however, the proposed models significantly exceeded this. FaceNet and ArcFace both achieved 99.61% and 99.82%, respectively, indicating that the proposed deep learning-based framework is not only accurate but also efficient in scenarios where privacy is a concern.

The proposed system demonstrated robust performance on the LFW dataset, renowned for its high variability and real-world complexity. The accuracy of the existing methods ranged from 95.45% to 96.67%, which is rather lower. The maximum accuracy was 95.72% for the proposed FaceNet model and 97.07% for ArcFace. On LFW, ArcFace outperformed most existing methods and remained competitive. In the proposed approach, the ArcFace model consistently achieved the highest accuracy and EER across all datasets examined-CASIA-3D, 105PinsFace, Faces94, and LFW-proving its superiority for privacy-preserving face recognition.

4.5 Computational cost analysis

Table 10 presents the computational cost of the proposed system when using FaceNet and ArcFace, evaluated with and without the LPP dimensionality-reduction step. The total cost is divided into four stages: feature extraction, LPP processing, encryption, and encrypted computation.

When LPP is applied, the feature vectors are reduced in size, which lowers the computational load in the encryption and encrypted computation stages. The LPP step takes 0.09873 s for both models. In this case, encryption requires 1.6570 s and encrypted computation takes 2.6947 s. The total processing time is therefore 4.49143 s for FaceNet and 4.45853 s for ArcFace. The small difference comes from the feature extraction stage, where ArcFace is slightly faster than FaceNet. Without LPP, the system processes full-dimensional embeddings, which increases the computational cost. Encryption time increases to 3.3141 s and encrypted computation increases to 5.3890 s. As a result, the total time becomes 8.7441 s for FaceNet and 8.7112 s for ArcFace, which is almost twice the time required when LPP is used. The results show that the LPP step significantly reduces the total computational cost of the system. Although ArcFace has slightly lower feature extraction time, the main factor affecting overall efficiency is the use of dimensionality reduction, which makes the system more suitable for real-time privacy-preserving applications.

4.6 Ablation study

In this study, two experimental pipelines were designed to evaluate the influence of embedding dimensionality on face recognition performance.

First, the baseline FaceNet model, which originally outputs a 128-dimensional embedding, was implemented and evaluated. To explore whether higher-dimensional representations could improve performance, the architecture was modified to produce a 256-dimensional embedding. The results obtained from both FaceNet-128D and FaceNet-256D are summarised in Table 11. However, these models did not yield satisfactory performance across the evaluated datasets. Consequently, a higher-capacity variant, FaceNet-512D, was adopted. From this model, a 512-dimensional embedding was extracted, and dimensionality reduction was applied to generate compact representations of 256D, 128D, 64D, and 32D. These reduced dimensions were chosen to facilitate efficient template storage and enable cryptographic operations in the subsequent secure system design.

Second, the baseline ArcFace model, which integrates a CNN backbone, embedding head, and margin-based loss function, was examined. Following the structure described in the original framework, the model was modified to extract 128D and 256D embeddings. The performance of these models, also reported in Table 11, did not meet the desired accuracy thresholds. To address this limitation, the standard ArcFace-512D embedding was utilised as the primary feature representation. Subsequently, Locality

Table 10 Computational cost of the proposed system for FaceNet and ArcFace

Model	FE time (s)	LPP (s)	Enc. time (s)	Comp. time (s)	Total time (s)
FaceNet	0.0410	0.09873	1.6570	2.6947	4.49143
ArcFace	0.0081	0.09873	1.6570	2.6947	4.45853
FaceNet	0.0410	–	3.3141	5.389	8.7441
ArcFace	0.0081	–	3.3141	5.389	8.7112

Table 11 Performance evaluation of FaceNet and ArcFace models on multiple datasets

Model	Dimension	Dataset	EER	Accuracy	Precision	Recall	F1-score
FaceNet	128D	CASIA-3D	0.205	79.34	0.0156	0.7967	0.0305
		PinsFace	0.448	57.92	0.0060	0.5238	0.0118
		Faces94	0.0313	96.84	0.1607	0.9691	0.2757
		LFW	0.4194	57.95	0.0009	0.5815	0.0018
FaceNet	256D	CASIA-3D	0.1711	82.86	0.0194	0.8293	0.0380
		PinsFace	0.4760	60.42	0.0052	0.4286	0.0103
		Faces94	0.0300	96.95	0.1658	0.9705	0.2832
		LFW	0.4214	66.36	0.0010	0.5006	0.0019
FaceNet	512D	CASIA-3D	0.040	98.51	90.87	95.12	92.94
		PinsFace	0.050	95.60	77.49	94.29	86.27
		Faces94	0.0059	99.61	94.48	99.41	96.89
		LFW	0.0652	95.72	91.91	93.32	92.61
ArcFace	128D	CASIA-3D	0.1139	88.61	0.0309	0.8862	0.0597
		PinsFace	0.4861	51.34	0.0051	0.5143	0.0100
		Faces94	0.0347	96.52	0.1479	0.9653	0.2565
		LFW	0.4389	56.15	0.0008	0.5597	0.0016
ArcFace	256D	CASIA-3D	0.1136	88.64	0.0310	0.8862	0.0599
		PinsFace	0.4686	53.22	0.0055	0.5333	0.0108
		Faces94	0.0335	96.67	0.1535	0.9664	0.2650
		LFW	0.4472	55.60	0.0008	0.5494	0.0016
ArcFace	512D	CASIA-3D	0.0082	99.02	96.80	99.17	97.97
		PinsFace	0.015	98.06	89.43	98.10	93.56
		Faces94	0.0059	99.82	91.03	99.41	94.99
		LFW	0.0344	97.10	91.27	96.53	93.83

Preserving Projections (LPP) were applied to reduce the 512D feature vector to 256D, 128D, 64D, and 32D, thereby enabling compact and discriminative representations suitable for secure matching and storage.

To assess the impact of encryption and dimensionality reduction (LPP) on recognition performance, an ablation study was conducted by comparing results with and without FHE using the CKKS scheme and with and without LPP, while keeping the reduced dimensionality fixed at 256 for the LPP-enabled configurations. The evaluation was performed across four datasets-CASIA-3D, 105PinsFace, Faces94, and LFW-using both FaceNet and ArcFace to clearly understand how each transformation affects verification performance.

In the Table 12, the behaviour of the models under plaintext and encrypted matching is shown for different LPP and FHE settings. For the CASIA-3D dataset, both FaceNet and ArcFace maintain strong performance across all conditions. ArcFace achieves a very low EER between 0.0082 and 0.0077, and its accuracy remains above 99% whether encryption is applied or not. FaceNet consistently achieves an EER of 0.040 across all cases, with only minor variations in accuracy. These results indicate that CKKS-based encryption has a negligible impact on verification accuracy in structured datasets.

In the 105PinsFace dataset, variations across settings are slightly more noticeable due to greater diversity in facial appearance. FaceNet alternates between an EER of 0.05 and 0.045, while ArcFace remains stable at 0.015 for all combinations of LPP and FHE. Despite these small shifts, both models maintain high accuracy within the 95–98% range, demonstrating that encrypted matching continues to perform reliably across more complex datasets.

Table 12 Ablation study showing the effect of LPP and FHE on the equal error rate (EER) across four datasets using FaceNet and ArcFace

Dataset	Model	LPP	FHE	EER
CASIA-3D	FaceNet	No	No	0.04
		Yes	No	0.04
		No	Yes	0.04
		Yes	Yes	0.04
CASIA-3D	ArcFace	No	No	0.0082
		Yes	No	0.0077
		No	Yes	0.0082
		Yes	Yes	0.0077
PinsFace	FaceNet	No	No	0.05
		Yes	No	0.045
		No	Yes	0.05
		Yes	Yes	0.045
PinsFace	ArcFace	No	No	0.015
		Yes	No	0.015
		No	Yes	0.015
		Yes	Yes	0.015
Faces94	FaceNet	No	No	0.0059
		Yes	No	0.0059
		No	Yes	0.0059
		Yes	Yes	0.0059
Faces94	ArcFace	No	No	0.0059
		Yes	No	0.0059
		No	Yes	0.0059
		Yes	Yes	0.0059
LFW	FaceNet	No	No	0.0652
		Yes	No	0.0650
		No	Yes	0.0652
		Yes	Yes	0.0650
Faces94	ArcFace	No	No	0.0344
		Yes	No	0.0340
		No	Yes	0.0344
		Yes	Yes	0.0340

In the Faces94 dataset, which contains clean and well-aligned face images, both models achieve nearly identical results across plaintext and encrypted scenarios. FaceNet maintains an EER of 0.0059 in all conditions, and ArcFace exhibits the same stable EER. The accuracy differences observed are extremely small, confirming that the encryption process does not affect performance in datasets with minimal intra-class variation.

The LFW dataset, which contains challenging real-world variations, shows slight differences across the models. FaceNet reaches an EER of around 0.065 with almost no change after encryption, while ArcFace remains around 0.034–0.0344 in all settings. These results show that even in uncontrolled environments, the encrypted system produces outcomes almost identical to those of the plaintext system. ArcFace proves more robust to such variations, maintaining lower EER values throughout.

Table 12 confirms that both LPP and CKKS-FHE preserve the discriminative structure of deep face embeddings. The EER remains effectively unchanged across all datasets, models, and settings, and accuracy variations remain minimal. These observations demonstrate that the proposed privacy-preserving system maintains strong recognition

quality while enabling secure encrypted matching, making it feasible and reliable for real-world face-recognition applications.

5 Security considerations

The proposed FR system's design aligns with ISO/IEC 24745, the global standard for protecting face information. This standard outlines the crucial needs to ensure that facial data is safely obtained, handled, stored, and transmitted while upholding user privacy and data integrity. Key aspects of compliance in the system embrace the irreversibility of stored templates, unlinkability, renewability, and protected processing.

Irreversibility is achieved by shrinking the original 512-dimensional features using LPP, followed by encryption using the CKKS FHE scheme, thereby confirming that the original face cannot be reconstructed. The system also facilitates unlinkability by storing encrypted, anonymised features unique to each application setting, effectively avoiding cross-matching across systems.

Renewability is enabled by the ability to regenerate templates using different projection matrices or encryption keys, without requiring re-enrolment of face data. All matching operations occur entirely in the encrypted domain, guaranteeing that raw or decrypted features are never exposed during handling.

Additionally, the system adheres to data minimisation and privacy-by-design principles by retaining only the most discriminating features through dimensionality reduction. These design selections collectively ensure substantial compliance with ISO/IEC 24745, making the system secure, privacy-protecting, and well-suited for deployment in sensitive and security-critical environments.

6 Conclusion

The proposed system presented a privacy-preserving face authentication system that integrates deep learning-based feature extraction, dimensionality reduction, and homomorphic encryption. The system was evaluated using two models-FaceNet and ArcFace-across multiple datasets. While both models were able to extract discriminative facial features, FaceNet consistently showed lower accuracy and higher error rates, particularly in complex scenarios. In contrast, ArcFace delivered superior performance, making it more suitable for reliable face authentication tasks. To enhance privacy and reduce computational cost, the system employed LPP to compress the 512-dimensional feature vectors before encryption. LPP effectively preserved local structure while reducing feature size, allowing efficient encrypted matching without significantly affecting accuracy. Future work can extend this system to multi-modal biometric fusion to improve accuracy and security. The system performance is analysed, but lightweight optimisation for edge devices is limited. So Future work can focus on improving efficiency for real-time deployment on resource-constrained hardware. Security evaluation mainly considers basic attack scenarios. Future work can include testing against advanced and stronger security threats to further improve system robustness.

Author contributions

This study was carried out in collaboration between the authors. Pavani Chitrapu was primarily responsible for the conception and design of the study, data collection, analysis, and manuscript drafting. Hemantha Kumar Kalluri provided critical revisions and intellectual guidance throughout the research process and supervised the findings of this work. Mahesh Kumar Morampudi provided critical revisions and intellectual guidance throughout the research process, and supervised the findings of this work. Satya Krishna Nunna provided critical revisions and intellectual guidance throughout the research process and supervised the findings of this work.

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Data availability
The datasets used in this study are available from their respective sources. The CASIA-3D FaceV1 dataset was obtained from the Institute of Automation, Chinese Academy of Sciences (CASIA). While the dataset is no longer openly downloadable from the previous CASIA portal, it remains available upon formal request to CASIA for academic research use. The 105Pins Face Dataset is publicly available at https://www.kaggle.com/datasets/hereisburak/pins-face-recognition. The Faces94 dataset is accessible via the University of Essex repository at http://cswww.essex.ac.uk/mv/allfaces/faces94.zip and community mirrors such as https://github.com/chenyijia1997/Face-datasets. The Labeled Faces in the Wild (LFW) deep-funneled dataset is publicly available at https://www.kaggle.com/datasets/danielbarus/lfw-deepfunneled. All datasets were used strictly for academic and non-commercial research purposes.

Declarations

Ethics approval and consent to participate
This study is based exclusively on publicly available datasets that were previously collected and released by other researchers. No new data were collected, and ethical approval was therefore not required. Not applicable. This study did not involve direct interaction with human participants and relied solely on publicly available data.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing financial or non-financial interests that could have influenced the work reported in this paper.

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